When Does an Image Edit Become Predictable? A Layer–Time Lens for Diffusion Editors

Anonymous submission

Abstract

Between reading "turn the car red" and painting it, what does a diffusion image editor represent? We build a tuned layer–time lens for a diffusion-transformer (DiT) image editor, decoding intermediate residual streams through the model's frozen output head and pair readings with controlled interventions. Under fixed underspecified recoloring prompts, early states predict realized outcomes while the frozen head still reads noise (car: 0.667 balanced accuracy vs. 0.333, $p{=}0.002$; natural-photo tree: hue MAE $55.9^{\circ}{\to}11.4^{\circ}$, $p{=}0.005$). As a held-out behavioral test, early best-of-five selection works on both architectures: recolor success rises from 20% to 67% on an independent 40-block editor ($p{=}4{\times}10^{-5}$), while on the natural-photo tree target error falls from 95.2° to 46.5° (mean rank 1.17, $p{=}3.7{\times}10^{-7}$). This code is *translated late*, in a sharp, timestep-invariant band (layers 52–54 of 60, width as narrow as one layer), into the velocity field that paints it. The boundary persists across tested edit families, an independent fine-tune, and a four-step distilled model. Transplant establishes sufficiency; steering shows that linear writability requires a spatial carrier. The lens also yields training-free levers: truncating classifier-free guidance after step two saves 45% of forward passes while preserving all tested edit families.

Introduction

Between reading "turn the car red" and painting it, what does a diffusion image editor actually do? The mechanics of diffusion suggest a gradual-development story, each of sixty blocks and twenty denoising steps contributing its small share – but this paper measures that story and finds it wrong. In the model we study, the realized edit is *predictable early* from an internal target-image code, and only *translated late*, in the last few blocks, into the flow field that does the painting.

The logit lens (nostalgebraist 2020) and tuned lens (Belrose et al. 2023) track a language model's next-token prediction by projecting intermediate residuals through its final unembedding. DiT image editors (Peebles and Xie 2023; Qwen Team 2025) have no analogous vocabulary: a patch hidden state is not an output distribution, so asking what a layer is drawing has required running the model to the end.

We build a layer–time lens for a production DiT editor, Qwen-Image-Edit-2511 (60 blocks), by decoding intermediate residual streams through the model's own frozen output head, and – crucially – pair passive readings with linear probes and controlled activation interventions (ablation, transplant, steering) that test what those activations can install or write.

Naively porting the logit-lens recipe reproduces the familiar mid-stack "nothing happens yet" appearance – and that appearance is itself misleading. With source and an underspecified recoloring instruction fixed across 48 seeds, a generation-level probe predicts the color ultimately chosen from layer 6, where the frozen head still reads noise. The outcome is held in a basis the head cannot parse until a narrow band near the top translates it into the velocity field that paints the result.

The main contributions of this work are as follows:

- **An instrument.** A layer–time instrument for a DiT image editor, combining a training-free frozen-head lens with tuned and differential variants, paired with linear probes and three intervention primitives (ablate, transplant, steer) with stated write semantics, under a falsification-first protocol – pre-registered predictions, per-image visual verdicts, and bit-exact gates.
- **A mechanism: the editor thinks with an internal image.** The realized edit is predictable early under a fixed underspecified instruction, carried as a target-image code, and translated into the velocity code only in a sharp, timestep-invariant band (layers 52–54, transition width one layer) that is consistent across the edit families tested and survives an independent fine-tune and a 4-step distillation. Transplant establishes sufficiency; steering and a rank ladder show that readable attributes require a spatial carrier and a variable-dependent write subspace.
- **Validation and a lever.** Held-out early best-of-five selection validates the readout on Qwen and an independent architecture, with about 68% measured compute reduction. Separately, truncating classifier-free guidance after step 2 saves 45% of forward passes across all tested edit families, suggesting that CFG's non-redundant role here is scope resolution rather than outcome selection.

This is exploratory, single-model-family work; every verdict was made on the images themselves, and we are explicit about where the lens is unreliable.

Related Work

Logit and tuned lenses. The logit lens (nostalgebraist 2020) projects intermediate residual streams through a language model's frozen unembedding; the tuned lens (Belrose et al. 2023) fits a small per-layer affine map instead. Both were built for autoregressive text models, where the unembedding gives every layer a shared, well-defined target vocabulary; a DiT editor has no such vocabulary; a patch's hidden state is not a distribution over anything until the whole network has run. We port both lens families to this setting – staying on the frozen-head side for honesty and crossing to the learned side only to render the internal image – with the linear representation hypothesis (Park, Choe, and Veitch 2024) licensing the informative disagreement between a frozen head and a probe.

Instruction-based diffusion image editing. Two lines established how a diffusion model can be made to follow an edit instruction at all. InstructPix2Pix (Brooks, Holynski, and Efros 2023) trains directly on synthesized instruction-image pairs so the model learns the edit end to end; Prompt-to-Prompt (Hertz et al. 2023) instead manipulates the model's cross-attention maps at inference time to inject the edit while preserving the source's structure, training-free. Both contribute a *mechanism for producing* the edit; neither asks what representation the network forms once an instruction is given, or where in the forward pass it becomes the pixels that change. Our lens targets that open question on exactly such a production editor – not a new editing mechanism, but a reading of the one it already has.

Thinking with images, and diffusion interpretability. A growing line has multimodal models reason *with* explicit intermediate images by externalizing a visual scratchpad (Hu et al. 2024; Wu et al. 2024; Li et al. 2025). We supply the mechanistic complement: an image that exists *latently* inside a generative editor before any output pixel is produced, with no explicit scratchpad or tool call involved. The Diffusion Lens (Toker et al. 2024) decodes a text-to-image model's *text encoder*; DiffLens and ICM (Shi et al. 2025; Zaleska et al. 2026) probe and steer internal features, including choices absent from an underspecified prompt; DreamReader unifies extraction, patching, steering, and stitching for text-to-image U-Nets (Sykora et al. 2026); and sparse autoencoders expose causal generation features (Surkov et al. 2025). Breaking the Lock-in finds early cross-seed convergence in text-to-image sampling and intervenes to increase diversity (Zhu et al. 2026). Probe-Select (Guo, Wei, and Jing 2026) predicts final quality from early activations; ELECT (Kim et al. 2025) scores early latent background consistency; and ADE-CoT (Qu et al. 2026) adds adaptive budgets and stopping at scale. We instead predict *which outcome* a fixed underspecified edit realizes and use selection only as a held-out behavioral test, while mapping how an image-editing DiT's realized outcome changes code across depth; we do not infer irreversible lock-in. We decode an editing DiT's own denoising stack as an image across both layer and time, building on the flow-matching and rectified-flow formulation (Lipman et al. 2023; Liu, Gong, and Liu 2023) and classifier-free guidance (Ho and Salimans 2022) that Qwen-Image-Edit-2511 (Qwen Team 2025; Peebles and Xie 2023) uses.

Causal interpretability and editing. Causal tracing / ROME (Meng et al. 2022) patches activations between runs, the move behind our transplant; Heimersheim and Nanda (2024) catalogue patching pitfalls that motivate our cautious interpretation of ablation. Recent analyses further show that detector-friendly diffusion features can produce artifacts when steered off distribution (Pegoraro, Johnson, and Nixon 2026), and that patching with multiple mediators cannot by itself isolate a unique causal pathway (Zhang, Hegde, and Zhang 2026). Accordingly, transplant is used only as a sufficiency test and steering as a bounded write test. Finally, activation addition (Turner et al. 2023) is the steering primitive we instantiate with a probe direction. All three were built for language models with a token vocabulary at every patched site; we port them to a continuous, spatial image stream, where sufficiency must be checked against a rendered image rather than a next-token distribution. Concept Lancet (Luo et al. 2025) already uses compositional representation transplant for diffusion editing; our transplant is therefore an interventional sufficiency check within a layer–time map, not a standalone editing-method claim. Our semantic readout uses CLIP (Radford et al. 2021) for lack of a better zero-shot alternative.

Proposed Method: A Layer–Time Lens and Intervention Probes

We study Qwen-Image-Edit-2511, a 60-block DiT editor whose image stream concatenates noisy target tokens with always-clean condition tokens (Figure 2). All quantities below are read from the *post-block residual* of selected blocks via forward hooks, captured in the model's native `bfloat16`: image-stream activations reach absmax $\sim 3\times 10^9$ even at layer 0, so a cast to `float16` silently overflows most elements to `inf` (bf16 capture is a lossless copy, not a downcast).

Four lenses. The *time lens* decodes the flow-matching clean-image estimate $\hat{x}_0 = x_t - \sigma_t v_\theta(x_t, t)$ through the VAE at step t: what the model currently believes it will output. The *depth lens* asks the analogous question across *depth*: it feeds an intermediate layer's hidden state through the model's own frozen final head (a non-affine `AdaLayerNormContinuous` + linear `proj_out`, bit-exact identical whether invoked at layer 59 or re-invoked earlier) and decodes the resulting pseudo-velocity to an $\hat{x}_0$ preview. Because that head's LayerNorm is non-affine, rescaling an intermediate state to the terminal norm is a no-op in principle and in practice (raw vs. rescaled agree to `max_rel_diff= 0`) – the *raw* head is the honest lens, with a standing self-consistency gate on every grid we report. The *tuned lens* (Belrose et al. 2023) crosses to the learned side for one purpose – rendering the mid-stack code back into an *image* at every depth: a per-(layer, timestep) ridge regression from the hidden state (3072-d) to

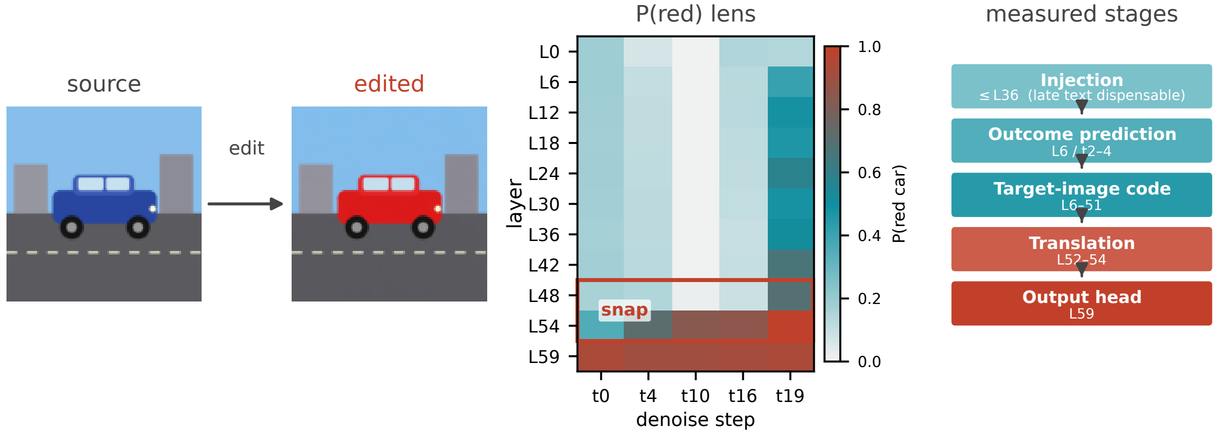

Figure 1: A layer–time lens for a DiT image editor. Decoding the frozen output head at intermediate layers of *car_red* gives a P(red car) grid across layer and denoising step (center); a sharp snap between layers 48 and 54 is where the head-readable signal first appears, while fixed-prompt generation-level probes show the realized outcome is already predictable from layer 6. We read this as five measured stages (right): injection early, an outcome-predictive target-image code through most of the stack, translation to the velocity code in a narrow band (L52–54), and readout at L59 – predictable early, translated late, not developed gradually.

the conditional-pass velocity (64-d), one dictionary per cell, $\hat{v}_{\ell,t} = W_{\ell,t}h + b_{\ell,t}$ solved in closed form after centering ($W_{\ell,t} = V^\top H(H^\top H + \lambda I)^{-1}$), with λ selected per cell on a held-out 10% *token* split – distinct from the zero-shot test below, which holds out an entire edit family and source image (full hyperparameters in the supplement).

A differential lens. The tuned lens renders the whole internal image; to isolate the *edit* from the reconstruction, we additionally difference it against a control. We run the same source and seed twice – once with the edit prompt, once with the literal instruction "keep the image unchanged" – and decode both through the same frozen dictionaries (no refit). Because the two runs share a seed, their step-0 latents are bit-identical, so $D_{\ell,t} = |\mathrm{decode}(h_\ell^{\mathrm{edit}}) - \mathrm{decode}(h_\ell^{\mathrm{unchanged}})|$ isolates the internal edit plan, the reconstruction cancelling by construction. A control-fidelity gate – the correlation between the no-op run and the untouched source – flags the rare case where the model cannot leave an image alone before any differential number is trusted.

Semantic readout. Neither lens is itself a score; each produces an image. We crop the edited region and score it zero-shot against a small bank of natural-language labels with CLIP (Radford et al. 2021), reporting the target-label probability. This is a whole-crop judgment, not a per-patch one, and inherits CLIP's failure modes on out-of-distribution previews; we gate on decode coherence and flag every violation.

Intervention probes. Reading a lens shows the representation is *consistent with* a hypothesis, not that it is *causal*. Three interventions, all forward-hook overwrites of the post-block residual, test distinct intervention-specific properties. **Ablate**: zero or mean-replace selected target rows at selected layers (an empty spec reproduces the baseline bit-for-bit); because repeated row replacement can be off-manifold, we use it only as a coarse stress test. **Transplant**: overwrite a recipient run's rows with a donor run's, matched by layer/step/patch – testing *sufficiency*. **Steer**: add a probe direction $\alpha\, d$ (in units of per-row RMS) without a donor pass – testing *linear writability*. All three are applied identically to both CFG passes to isolate the intervention rather than a lopsided guidance combination.

Protocol: predict, look, gate. Three rules make the *reasoning* reproducible, not just the runs. **Predict, then run**: every intervention was pre-registered; refutations are reported as prominently as confirmations, and several headline results are pre-registered predictions that failed informatively. **Look at every image**: no verdict rests on CLIP alone – every run carries a per-image visual verdict recorded before the aggregates were read, and where the two disagree the image wins. **Gate the instrument**: the empty-ablation and raw-vs-rescaled checks are standing regressions re-run per experiment. Porting the method to another editor needs only a native-dtype residual hook, the frozen-head gate, and region-restricted probes – the recipe executed unchanged in the cross-model replication below.

Experiments

Experimental Setup

We evaluate on Qwen-Image-Edit-2511, a 60-block DiT editor, on edit batteries spanning five categories – color, background, style, removal, and addition – over natural photographs and one flat-art source (10 cases for the time-lens saturation sweep and the preview lever, 20 for the IoU-based onset and CFG-truncation checks); every quantitative num-

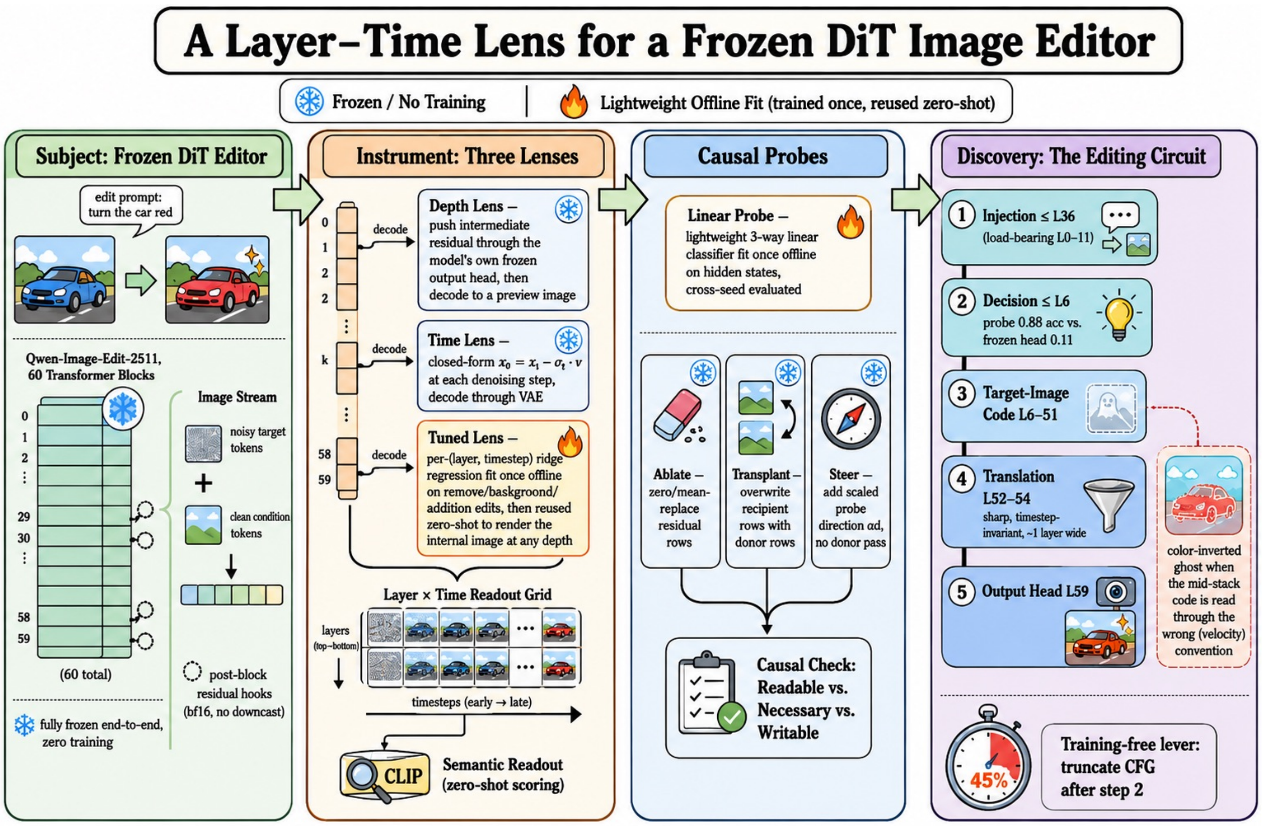

Figure 2: Pipeline. The image stream concatenates noisy target tokens with always-clean condition tokens; each block's post-block residual is tapped and read two ways – the *depth lens* pushes it through the model's own frozen output head before decoding, the *time lens* instead reads the current step's head output combined with the current latent ($\hat{x}_0 = x_t - \sigma_t v$). Both routes end in a decode-then-CLIP readout, computed identically for both true-CFG passes.

ber is paired with a per-image visual verdict recorded before the aggregate is read, per the predict-look-gate protocol above. Three readouts do the reporting: the frozen output head's decode, scored zero-shot with CLIP against a small label bank; cross-seed probes, including a 48-seed fixed-prompt outcome probe whose independent unit is one generation; and, for the tuned lens, held-out R^2 against the true conditional-pass velocity. Two further checkpoints sharing Qwen-Image-Edit-2511's transformer classes – FireRed-Image-Edit-1.1, an independent fine-tune, and Rapid-AIO, a 4-step distilled merge – let us ask whether a finding persists beyond one training run (tested below).

The Realized Edit Is Predictable Early

In time. If the edit surfaced progressively, the readout would saturate late and at roughly the same rate. Running the time lens over a battery of 10 cases across 5 edit categories, the step at which P(target) first reaches within 0.05 of its final value is instead *edit-dependent* and mostly *immediate*: color, background, and style edits reach the criterion at step 0–2, and only a wholesale new-object insertion (*add_boat*) does so as late as step 7. A finer IoU-mask-based onset measurement over 20 cases shows the split is categorical, not continuous: 18 of 20 – spanning final-edit-area fraction 0.003 to 0.99 – onset at step 0 regardless of how large or novel-looking the finished edit is, and no continuous proxy predicts the delay (edge-novelty ρ=0.24, p=0.30; area ρ=−0.44, running the *wrong* way). Only the two edits that insert genuinely novel structure with no existing pixels to build on – *add_boat* and *add_runner* – start late.

In depth, the naive lens says "late". The frozen depth lens tells a different-looking story. Decoding *car_red*'s intermediate layers through the model's own head, the edit becomes head-readable only in a sharp snap between layers 48 and 54 of 60 – exactly the familiar mid-stack "nothing happens yet" appearance of an LLM logit lens.

But the realized outcome is predictable 40 layers earlier. An explicit-prompt probe first reveals the basis mismatch: predicting which of three colors was requested reaches 0.883 accuracy at **layer 6**, step 4, where the frozen head reads only P(red car)=0.111 (Figure 3). To separate outcome prediction from reading a color word, we hold both source and the underspecified prompt "change the car to a different color" fixed across 48 new seeds and label each final image post hoc. One mean-pooled vector per generation predicts the realized color at L6/step 4 with 0.667 balanced accuracy (chance 0.333; 500-permutation $p{=}0.002$). A continuous decoder at L6/step 2 reduces final-hue error from $22.3°$ to $8.3°$ ($R^2{=}0.815$). Thus the outcome, not merely the instruction, is linearly predictable early; this does not claim irreversible commitment. A pre-registered real-photo follow-up failed its discrete gates (mug class collapse; heterochromatic eyes), but mug continuous hue replicated ($17.0°{\to}10.7°$, $p{=}0.005$; L59 $16.7°$), while eyes were null ($p{=}0.796$). A separate tree replication passed 48/48: $55.9°{\to}11.4°$ ($p{=}0.005$), with L59 at $51.1°$. On the independent 40-block Boogu architecture, L4/step 2 similarly gives $39.9°{\to}6.9°$ ($p{=}0.005$), though L39 is stronger and 35/48 retain source blue. Turning prediction into held-out selection, a Boogu selector chooses a recolor in 8/12 new five-seed groups versus 20% random ($p{=}4.1{\times}10^{-5}$), all eight solvable; L4/step 0 gives 4/12 ($p{=}0.188$). Two separately pre-registered Qwen target tasks pass. On 12 tree groups, L6/step 2 cuts error from $95.2°$ (random) to $46.5°$ (oracle $43.2°$; mean rank 1.17, $p{=}3.7{\times}10^{-7}$), while step 0 is null ($p{=}0.381$). A current-environment car replication cuts $25.7°$ to $11.1°$ (oracle $8.7°$; mean rank 2.00, $p{=}0.0086$); its step-0 control retains partial signal but misses the frozen rank and p gates. Actual interrupted captures and winner reruns save about 68% versus full best-of-five. This selects the closest candidate; some groups contain no target-like hue. Early prediction transfers, but the Qwen terminal drop and translation do not.

Where the instruction enters. A complementary text-stream ablation locates where the prompt is *consumed*: zeroing the model's text tokens in only the first 11 blocks garbles the whole scene, while zeroing them after layer 36 barely dents the edit ($P(\text{red}){=}0.993$). The instruction is fused into the image stream early; for this color edit, later text tokens are dispensable by layer 36 – the "injection" stage of the circuit in Figure 1, whose full text-band table is in the supplement. Together with the probe, this gives the paper's five-stage picture: injection ($\leq$L36), early outcome prediction, a target-image code, translation (L52–54), and the output head.

The Edit Is Translated Late: Two Codes

If the realized outcome is predictable early but the frozen head cannot read it until layer 52, the mid-stack must hold the edit in a *different code*. Three measurements show what that code is and where it is converted.

Two decoding conventions cross over. At an intermediate layer the frozen head yields a per-token vector v_ℓ we normally treat as a velocity, forming $\hat{x}_0 = x_t - \sigma_t v_\ell$ (variant A). Decoding v_ℓ instead *directly*, as if it were already a clean latent (variant C), inverts the story: for *car_red* at step 4, at L24 the standard decode A scores $P(\text{red}){=}0.12$ (still wrong-colored) while the direct decode C already scores 0.66; at L59 this reverses (A = 0.93, C = 0.02). The mid-stack is not blank – it encodes the target content in an *image-like* convention, and a narrow band near the top reconciles it to the velocity convention the head consumes. (This is a whole-vector reweighting, not a literal sign flip: essentially no individual channel of v_ℓ has a negative slope against v_{59}, mean 0.003.)

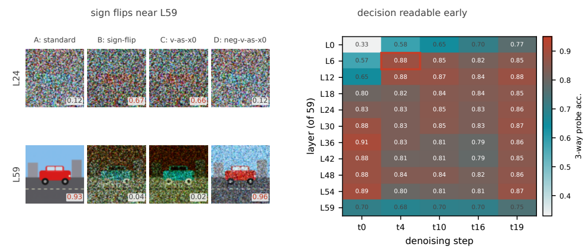

Figure 3: Left: four ways of turning an intermediate layer's frozen-head output into an image for *car_red* at $t{=}4$ (A: standard $\hat{x}_0{=}x_t{-}\sigma_t v_\ell$; B: sign-flipped; C: v_ℓ decoded directly; D: $-v_\ell$ decoded directly) – at L24 the standard decode is wrong-colored while the sign-flipped/direct decodes already look red, and at L59 this reverses. Right: cross-seed 3-way probe accuracy (chance = 0.33) by layer/step; the headline cell L6/t4 (0.883) is far above the frozen-head reading at the same cell, and probe accuracy drops at L59 relative to L48–L54.

The conversion is a sharp, timestep-invariant band. A dense every-layer sweep from L44 to L59 (Figure 5) locates the A/C crossover precisely: transition midpoint L53.6 at step 16 (width one layer) and L52.7 at step 4, the two differing by under one layer – the boundary's *location* is timestep-invariant. Crucially, the cross-seed probe accuracy is close to *flat* (0.77–0.84) across every layer from L44 through L58 and drops only at the terminal layer L59 (0.68–0.70). So the translation that changes what the head can read (L52–54) and the loss of linear separability (L59) are two *distinct* events sharing a neighborhood: the target-image code survives the translation intact and is degraded only by the final, velocity-specific projection.

A tuned lens renders the internal image, layer by layer. To confirm the mid-stack code is a genuine *image* rather than a probe artifact, we fit a per-(layer, timestep) tuned lens, as described above, on *remove*/*background*/*addition* edits and evaluate it *zero-shot* on the held-out *car_red* color family. Held-out R^2 rises with depth from 0.73–0.89 at L6 to 0.998 at L59, where a standing identity gate confirms the target is correct (decoding L59 through the frozen head reproduces the true velocity bit-for-bit, $\max|\text{diff}|{=}0$). Every depth down to L6 decodes a coherent scene: a pre-registered prediction that readability itself would have a boundary is refuted – the L52–54 boundary is a property of the frozen head's one fixed projection, not of the representation. At step 0 the pixels are pure noise, so the decode comes entirely from the velocity code, and we watch the edit enter the internal image directly (Figure 4): the source's blue car persists through L24, turns orange-red by L36, the scene composes by L44, and the car is fully red by L52. The same frozen dictionaries transfer with zero retraining to six further held-out cases (style, more colors, removal, background, addition); *add_runner* even reproduces the delayed onset on the internal image's own time axis – an empty path at step 0, a clear runner by step 4.

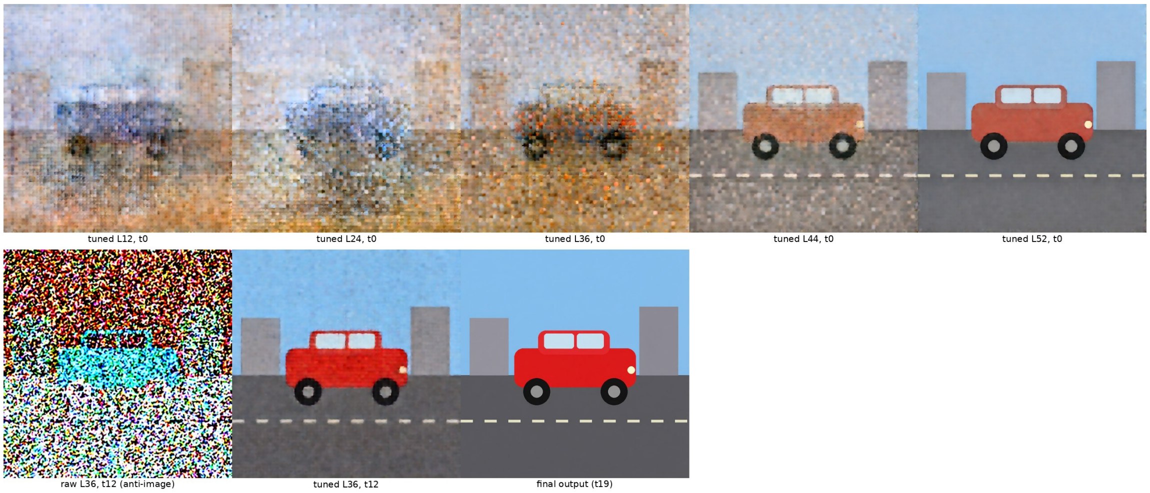

Figure 4: A tuned lens renders the internal image the frozen head cannot read. Top: per-(layer, timestep) ridge translators, fit on *remove/background/addition* edits and evaluated zero-shot on the held-out *car_red* color edit, decode a coherent scene at every depth at $t=0$ – the edit enters the internal image layer by layer. Bottom: at $t=12$ the raw frozen head renders the identical L36 hidden state as a color-inverted ghost, while the tuned lens at the same cell is already coherent.

The anti-image, mechanized. This also explains a striking artifact: the mid-stack *frozen*-head decode at later steps is a color-*inverted* ghost (a cyan car under a red-brown sky) – the image-like mid-stack code forced through a head built to consume a velocity, since $\hat{x}_0 = x_t - \sigma v$ flips the sign of an image-like quantity. The tuned lens at the same cell is already coherent.

Transplantable, Writable, and Carrier-Bound

Reading a lens shows only that a representation is *consistent with* an edit, not what it can install. Whole-token ablation across many layers drives the edited region off-manifold, so we use it only as a coarse stress test rather than evidence for fine-grained necessity.

Transplant establishes sufficiency. Transplanting a *car_red* donor's car-region hidden states into a *car_green* recipient (which alone scores $P(\text{red}){=}0.011$) flips the output to red from *any* of four donor bands, including **L48–59** (0.940). The donor representation is therefore sufficient to install the edit. A content control confirms specificity: splicing the same donor's *background*-region tokens (right band, wrong content) yields a garbled "blue car" (0.991), not red.

Readability does not imply unconstrained writability. A single stored *direction* – the probe's d_{red}, no donor pass – added to the recipient's car rows gives a sharp sigmoidal dose-response (0.024 at $\alpha{=}0.5$, 0.977 at $\alpha{=}1$, saturating to $\alpha{=}8$); a random direction of equal norm is inert (0.017), and the negated direction overshoots to blue (0.945), reproducing the anti-image under an active intervention. But writability is bounded: at effective doses the steered region is not a recolored car but a flat red patch that *replaces* the car's geometry – the direction installs *which* hue, not *where to stop*. Supplying that missing "where" with the edited object's own silhouette mask turns every previously-failing dose into a clean in-place recolor. And conjuring an *object* outruns any single direction: the identity probe direction is inert (the read and write subspaces differ), while a rank ladder over the donor−recipient difference resolves identity coarse-to-fine – category at rank 1, a coherent instance by rank 4, donor-faithful by rank 64 (2% of the model dimension). The read and write subspaces differ, and a writable attribute still requires a spatial carrier.

A Preview Lever: The Differential Lens at 15% NFE

The differential lens (*Proposed Method*) isolates the internal edit plan from the reconstruction. It first settles a caveat left open above: token-level R^2 *inside* the edit region is 0.88–0.97, matching the full-image value, so the tuned dictionaries carry the edit-specific signal itself, not incidental scene detail.

Content is early; separation is gradual. A pre-registered prediction that D would already localize tightly at every step is refuted, informatively. The edit *content* is present at step 0 (the tuned lens already shows red eyes, a black trunk), but the *differential* at step 0 is a diffuse whole-frame field – the edit prompt perturbs the entire velocity relative to the no-op, not just the edited region – and condenses onto the edit's footprint only gradually (localization of D's top mass at L52 climbs monotonically across steps, e.g. $0.21{\to}0.47{\to}0.67{\to}0.83$ for a cat's-eyes recolor). *Present* (content in the internal image) and *separated* (the differential has found where it lives) are different quantities. The no-op control also caught one flat-art scene that *cannot* "do nothing" (it renders a transparency checkerboard, and is discarded), motivating the control-fidelity gate applied before any differential number is trusted.

A preview lever. Because the differential carries the edit's footprint, we ask whether it does so early enough to be useful.

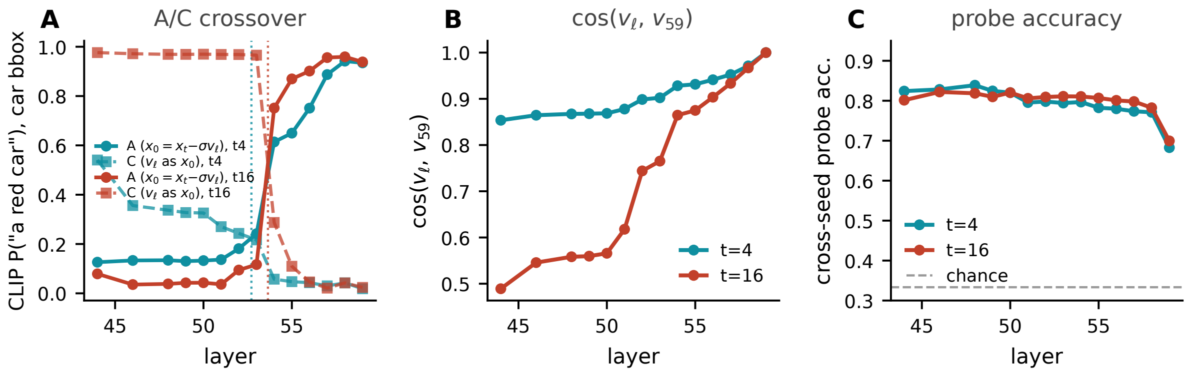

Figure 5: A dense, every-layer sweep of *car_red* over L44–L59 sharpens the coarse L48/L54 snap to a precise boundary. **A**: the standard decode and the direct-v decode cross between L52 and L54 at both steps shown. **B**: cosine similarity of the intermediate pseudo-velocity to the terminal v_{59} rises sharply in the same band. **C**: cross-seed probe accuracy is flat from L44 through L58 and drops only at the terminal layer.

At step 2 – the third of 20, ≈15% of the forward passes – across a 10-case, five-family battery of natural photographs (all passing the control-fidelity gate, $r \in [0.86, 0.99]$), the median IoU between $D_{\mathrm{L52},t2}$'s top mass and the true edit region is 0.33, 6.6× chance, and all 10 cases sharpen from step 0 to step 2 (figure in the supplement): an object slated for removal lights up as a ghost, a keep-vs-replace background segmentation emerges unsupervised, and a tiny recolor glows at its exact site. Because the no-op control is *independent of the edit prompt*, it is computed once per source image and cached; previewing a *candidate* edit then costs three forward passes plus one frozen-dictionary decode, not a full 20-step run – a training-free way to triage many prompts against one image before running any to completion.

Generalization Across Checkpoints

Most mechanistic findings so far concern one architecture. We re-ran the depth-lens crossover and cross-seed probe on the two further checkpoints from *Setup* (FireRed-Image-Edit-1.1 and Rapid-AIO). **The translation boundary and explicit-prompt early readability replicate.** FireRed's A/C crossover lands at L53–56, Rapid-AIO's at L52–54 – both on Qwen-Image-Edit-2511's L52–54 band; FireRed's probe reads 0.852 at L6 and dips to 0.69 at L59, the same flat-then-dip shape. **The readable direction itself transfers across models**: a probe trained on one model and tested on the *other* reaches 0.84/0.92 accuracy, and the two models' L12 probe directions have cosine similarity 0.833 – the edit lives in nearly the same direction of the shared 3072-d space, not merely the same layer index. The full tuned-lens toolchain also ports to FireRed (deep-layer R^2 0.97–0.999, anti-image reproduced).

A Training-Free Lever: Truncating Classifier-Free Guidance

The layer–time map suggests a simple existence proof: classifier-free guidance may not need both forward passes at all 20 steps. Running the unconditional pass only for the first two steps and the conditional pass alone thereafter leaves all four tested edit families visually intact at a **45% reduction in forward passes** (a 20-case battery confirms it, mean 1.77× speedup). Dropping CFG entirely suggests where its non-redundant role lies in these cases: recolor, removal, and addition still land from the conditional pass alone, but background swap composites the new scene *around* un-erased source structure ($P(\text{beach})$=0.45 vs. ≥0.996) – CFG's non-redundant contribution here is consistent with resolving *scope*, not selecting edit content. The same layer–time structure survives 7.7× step distillation (20/20 cases land), so distillation compresses the sampling trajectory but not the structure the lens exposes.

Routing conservation explains coherent transplant. A RESTORE-inspired five-seed audit (Cho et al. 2026) finds destructive replacement more routing-distorting than transplant/red steering in 5/5 seeds (contrast 0.0356, exact p=0.03125). Matched-donor TV is 0.0031 with geometry 5/5; background-donor TV is 0.0677 with geometry 0/5. Red steering writes red 5/5 but preserves geometry 0/5, matching the detection–intervention gap (Pegoraro, Johnson, and Nixon 2026); this supports tested coherence only.

Limitations

Translation/intervention evidence uses Qwen plus two same-lineage checkpoints; Boogu supports only outcome prediction/selection. The fixed-prompt probe has one synthetic source; natural-image follow-ups give two replications and one unsuitable eye case. Selection has 12 groups per task and cannot create absent targets. One-way L6 hue transfer (mug→tree p=0.002; reverse p=0.968) precludes an object-invariant subspace. Because CLIP can prefer a flat patch to a true recolor, quantitative claims carry pre-registered visual verdicts. We claim tested outcome predictability, transplant sufficiency, and carrier-bounded writability, not irreversible commitment or necessity. In summary, the lens predicts outcomes and Qwen's late translation; selection spans two architectures and routing explains coherent transplant.

References

Belrose, N.; Furman, Z.; Smith, L.; Halawi, D.; Ostrovsky, I.; McKinney, L.; Biderman, S.; and Steinhardt, J. 2023. Eliciting Latent Predictions from Transformers with the Tuned Lens. *arXiv preprint arXiv:2303.08112*.

Brooks, T.; Holynski, A.; and Efros, A. A. 2023. InstructPix2Pix: Learning to Follow Image Editing Instructions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Cho, H.; Baek, D.; Kim, Y.; and Ham, B. 2026. Improving Visual Token Reduction via Rectifying Distortions for Efficient Multimodal LLM Inference. *arXiv preprint arXiv:2606.01711*. Accepted at ICML 2026.

Guo, H.; Wei, H.; and Jing, B. 2026. Toward Early Quality Assessment of Text-to-Image Diffusion Models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13186–13195.

Heimersheim, S.; and Nanda, N. 2024. How to Use and Interpret Activation Patching. *arXiv preprint arXiv:2404.15255*.

Hertz, A.; Mokady, R.; Tenenbaum, J.; Aberman, K.; Pritch, Y.; and Cohen-Or, D. 2023. Prompt-to-Prompt Image Editing with Cross-Attention Control. In *International Conference on Learning Representations (ICLR)*.

Ho, J.; and Salimans, T. 2022. Classifier-Free Diffusion Guidance. *arXiv preprint arXiv:2207.12598*.

Hu, Y.; Shi, W.; Fu, X.; Roth, D.; Ostendorf, M.; Zettlemoyer, L.; Smith, N. A.; and Krishna, R. 2024. Visual Sketchpad: Sketching as a Visual Chain of Thought for Multimodal Language Models. In *Advances in Neural Information Processing Systems*.

Kim, J.; Lee, S.; Kim, N.; and Lee, S. 2025. ELECT: Early Latent Consistency for Zero-Shot Candidate Selection in Instruction-Guided Image Editing. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 8209–8218.

Li, C.; Wu, W.; Zhang, H.; Xia, Y.; Mao, S.; Dong, L.; Vulić, I.; and Wei, F. 2025. Imagine while Reasoning in Space: Multimodal Visualization-of-Thought. *arXiv preprint arXiv:2501.07542*.

Lipman, Y.; Chen, R. T. Q.; Ben-Hamu, H.; Nickel, M.; and Le, M. 2023. Flow Matching for Generative Modeling. In *International Conference on Learning Representations (ICLR)*.

Liu, X.; Gong, C.; and Liu, Q. 2023. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. In *International Conference on Learning Representations (ICLR)*.

Luo, J.; Ding, T.; Chan, K. H. R.; Min, H.; Callison-Burch, C.; and Vidal, R. 2025. Concept Lancet: Image Editing with Compositional Representation Transplant. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 28502–28512.

Meng, K.; Bau, D.; Andonian, A.; and Belinkov, Y. 2022. Locating and Editing Factual Associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35.

nostalgebraist. 2020. Interpreting GPT: The Logit Lens. LessWrong. Accessed 2026.

Park, K.; Choe, Y. J.; and Veitch, V. 2024. The Linear Representation Hypothesis and the Geometry of Large Language Models. In *International Conference on Machine Learning (ICML)*.

Peebles, W.; and Xie, S. 2023. Scalable Diffusion Models with Transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*.

Pegoraro, A.; Johnson, D. D.; and Nixon, J. 2026. Look But Don't Touch: Visual Features in Diffusion Models Are Easy to Detect but Hard to Steer. *arXiv preprint arXiv:2606.31699*.

Qu, X.; Yuan, Z.; Tang, J.; Chen, R.; Tang, D.; Yu, M.; Sun, L.; Bai, Y.; Chu, X.; Gou, G.; Xiong, G.; and Cai, Y. 2026. From Scale to Speed: Adaptive Test-Time Scaling for Image Editing. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 23272–23282.

Qwen Team. 2025. Qwen-Image Technical Report. *arXiv preprint arXiv:2508.02324*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *International Conference on Machine Learning (ICML)*, 8748–8763.

Shi, Y.; Li, C.; Wang, Y.; Zhao, Y.; Pang, A.; Yang, S.; Yu, J.; and Ren, K. 2025. Dissecting and Mitigating Diffusion Bias via Mechanistic Interpretability. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8192–8202.

Surkov, V.; Wendler, C.; Mari, A.; Terekhov, M.; Deschenaux, J.; West, R.; Gulcehre, C.; and Bau, D. 2025. One-Step is Enough: Sparse Autoencoders for Text-to-Image Diffusion Models. In *Advances in Neural Information Processing Systems*.

Sykora, Q.; Williams, F.; Hu, Y.; Duan, P.; Zhou, B.; Wetzstein, G.; and Aigerman, N. 2026. DreamReader: Interpreting and Editing Diffusion Models via Representation Extraction. *arXiv preprint arXiv:2603.13299*.

Toker, M.; Orgad, H.; Ventura, M.; Arad, D.; and Belinkov, Y. 2024. Diffusion Lens: Interpreting Text Encoders in Text-to-Image Pipelines. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 9713–9728.

Turner, A. M.; Thiergart, L.; Leech, G.; Udell, D.; Vazquez, J. J.; Mini, U.; and MacDiarmid, M. 2023. Activation Addition: Steering Language Models Without Optimization. *arXiv preprint arXiv:2308.10248*.

Wu, W.; Mao, S.; Zhang, Y.; Xia, Y.; Dong, L.; Cui, L.; and Wei, F. 2024. Mind's Eye of LLMs: Visualization-of-Thought Elicits Spatial Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems*.

Zaleska, K.; Popek, Ł.; Wysoczańska, M.; and Deja, K. 2026. Attention, May I Have Your Decision? Localizing Generative Choices in Diffusion Models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9818–9828.

Zhang, J.; Hegde, C.; and Zhang, Y. 2026. The Curse of Multiple Mediators in Activation Patching. *arXiv preprint arXiv:2606.20945*.

Zhu, Y.; Chen, R.; Lin, W.; Zhang, J.; and Qi, Y. 2026. Breaking the Lock-in: Diversifying Text-to-Image Generation via Representation Modulation. In *International Conference on Machine Learning (ICML)*.

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